

Answer to Question 5

Neural Network Classification for Disease Prediction

Question

In the context of Neural Networks for classification, consider a dataset used to predict whether a patient has a disease based on attributes such as Age, Blood Pressure, Glucose Level, and Lifestyle Risk. The output indicates whether the disease is present (Yes) or not (No). Using this dataset, explain how a neural network model can be trained using backpropagation to classify the data. Also, evaluate the model performance by comparing actual and predicted outputs. Using the dataset, compute the values of True Positives, True Negatives, False Positives, and False Negatives. Based on these values, analyze the performance of the neural network using suitable evaluation metrics. Also, explain how preprocessing and feature selection improve the model's accuracy and reliability. (5 Marks)

1. Overview of the Problem

This is a binary classification problem where a neural network must predict whether a patient has a disease (Yes/No) using four input features: Age, Blood Pressure, Glucose Level, and Lifestyle Risk. Categorical attributes (Blood Pressure, Glucose Level, Lifestyle Risk) are encoded numerically, for example Low = 0, Medium/Normal = 1, High = 2, so that the network can process them mathematically. The target output (Yes/No) is also encoded as 1/0.

2. Neural Network Architecture

A simple Multilayer Perceptron (MLP) can be used:

- **Input Layer:** 4 neurons, one for each feature (Age, Blood Pressure, Glucose Level, Lifestyle Risk).
- **Hidden Layer(s):** One or two hidden layers with neurons using an activation function such as ReLU or Sigmoid to capture non-linear relationships between symptoms and disease risk.
- **Output Layer:** 1 neuron with a Sigmoid activation function, producing a probability between 0 and 1. A threshold of 0.5 is used to classify the output as Yes (≥ 0.5) or No (< 0.5).

3. Training the Model Using Backpropagation

- **Forward Pass:** For each training example, the input features (Age, BP, Glucose, Lifestyle Risk) are passed through the network. Each neuron computes a weighted sum of its inputs plus a bias, which is passed through an activation function to produce an output. This continues layer by layer until the final predicted output (probability of disease) is generated.
- **Error/Loss Calculation:** The predicted output is compared with the actual output (Yes/No) using a loss function such as Binary Cross-Entropy Loss. This measures how far the prediction is from the true label.
- **Backward Pass (Backpropagation):** The error is propagated backward from the output layer to the hidden and input layers using the chain rule of calculus. The gradient of the loss with respect to each weight is calculated, indicating how much each weight contributed to the error.
- **Weight Update:** Weights and biases are updated in the direction that reduces the error, using an optimization rule such as Gradient Descent: $w_{new} = w_{old} - (learning\ rate) \times (gradient\ of\ loss\ with\ respect\ to\ w)$.

- **Iteration:** This forward-backward cycle is repeated over many epochs across the training data (examples 1-5, or a larger training set) until the loss becomes minimal and the network converges to stable weights that classify patients accurately.

4. Computing TP, TN, FP, FN from the Dataset

Comparing the Actual Output with the Predicted Output for each example:

Example	Actual Output	Predicted Output	Result
1	Yes	Yes	True Positive (TP)
2	Yes	No	False Negative (FN)
3	No	No	True Negative (TN)
4	Yes	Yes	True Positive (TP)
5	No	Yes	False Positive (FP)

Therefore:

TP (True Positives)	2 (Examples 1, 4)
TN (True Negatives)	1 (Example 3)
FP (False Positives)	1 (Example 5)
FN (False Negatives)	1 (Example 2)

5. Evaluation Metrics

Using TP = 2, TN = 1, FP = 1, FN = 1, and Total examples = 5:

- **Accuracy** = $(TP + TN) / \text{Total} = (2 + 1) / 5 = 3/5 = \mathbf{60\%}$ — the model correctly classified 3 out of 5 patients.
- **Precision** = $TP / (TP + FP) = 2 / (2 + 1) = \mathbf{66.7\%}$ — of all patients predicted as having the disease, 66.7% actually had it.
- **Recall (Sensitivity)** = $TP / (TP + FN) = 2 / (2 + 1) = \mathbf{66.7\%}$ — of all patients who actually had the disease, the model correctly identified 66.7% of them.
- **F1-Score** = $2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall}) = \mathbf{66.7\%}$ — the harmonic mean of precision and recall, useful when the two need to be balanced.

Analysis: The model shows moderate performance (60% accuracy). Example 5 was misclassified as a False Positive, and Example 2 was misclassified as a False Negative. In a medical diagnosis context, False Negatives are especially critical since a diseased patient (Example 2) is predicted as healthy, which can delay treatment. This indicates the model needs improvement, for example by using more training data, tuning the learning rate, or adjusting the classification threshold to reduce missed diagnoses.

6. Role of Preprocessing and Feature Selection

Data Preprocessing improves accuracy and reliability by:

- **Normalization/Standardization:** Scaling Age and Glucose Level to a common range (e.g., 0-1) so that no single feature dominates the learning process due to its larger numeric range.

- **Encoding categorical data:** Converting Blood Pressure, Glucose Level, and Lifestyle Risk (Low/Medium/High) into numeric values so the neural network can process them.
- **Handling missing/noisy data:** Removing or imputing missing values so the model learns from clean, consistent data.

Feature Selection improves the model by:

- Identifying the attributes (e.g., Glucose Level, Blood Pressure) that are most strongly correlated with the disease outcome, and removing irrelevant or redundant features.
- Reducing model complexity and overfitting, which improves generalization to new/unseen patients.
- Speeding up training and making the model more interpretable for medical practitioners.

Together, proper preprocessing and feature selection lead to faster convergence during backpropagation, fewer misclassifications (reducing FP/FN counts), and a more accurate and reliable disease-prediction system.

(5 Marks)